

Lecture 03 - Exercises

These exercises accompany Lecture 03, “Describing and Summarizing Data.” They practise the full descriptive-statistics workflow: identifying variables, choosing appropriate summaries, moving from raw data to tables and graphs, computing measures of location and dispersion, transforming prices into returns, and interpreting all results in context.

The exercises are in **two parts, and the order is deliberate.**

Part 1 is done by hand on small datasets. The point is not the arithmetic itself, but the mechanics behind each formula: what a weighted mean actually averages, why the sample variance divides by $n-1$, how compounding differs from simple addition, and why log returns add over time while simple returns multiply. Once you have computed these things by hand at least once, the numbers that Python later prints become meaningful rather than mysterious. Work Part 1 in written sentences and show the main calculations.

Part 2 is done in Python, on larger and real datasets where hand computation is impractical. The logic is identical to Part 1 — the same summaries, the same conventions, the same interpretation discipline — but now applied at a scale where a computer is not a convenience but a necessity. You already have the `numpy`, `pandas`, and `matplotlib.pyplot` skills from Lectures 01 and 02, so Part 2 assumes them.

A few parts are marked (*stretch*); treat these as optional extensions if you are short on time. **Full solutions for both parts are provided together in a single notebook.**

Part 1 - Working by Hand

Answer in clear written sentences and show the main calculations. Do not use a computer for this part; the goal is to understand the mechanics.

Exercise 1 - Dataset Vocabulary

For each situation, identify the population, sample, observation, and at least two variables.

1. A bank analyzes 5,000 mortgage applications submitted last year to understand default risk.
2. A portfolio manager studies daily adjusted closing prices for Apple from 2012-01-03 to 2021-10-01.
3. An online retailer studies 60 invoices from one month to summarize customer purchasing behavior.

Then:

4. For each situation, explain one parameter that the analyst might care about and one statistic that could be computed from the sample.
5. For situation 2, name an estimator you could use for that parameter, and explain in one or two sentences the difference between the *estimator* (the rule) and the *estimate* (the number it produces on this particular sample).

Exercise 2 - Variable Types and Meaningful Summaries

Classify each variable as nominal categorical, ordinal categorical, binary, discrete numerical, or continuous numerical. Then name one meaningful summary and one inappropriate summary for that variable.

1. Firm sector.

2. Credit rating: AAA, AA, A, BBB, BB.
3. Whether a borrower defaulted.
4. Invoice amount in euros.
5. Number of items bought.
6. Daily simple return.
7. Customer satisfaction: very low, low, medium, high, very high.
8. Invoice ID.

Exercise 3 - From Raw Labels to a Frequency Table

A satisfaction variable (on the scale very low, low, medium, high, very high) contains the following 20 observations:

medium, low, high, medium, very high, low, medium, high, low, medium,
high, very low, medium, low, high, medium, very low, high, medium, very high

1. Create a frequency table, keeping the categories in their natural order.
2. Create a relative frequency table.
3. Identify the mode.
4. Choose an appropriate graph and explain why it is appropriate, and why the order of the categories must be preserved.
5. Write two sentences interpreting the distribution.

Exercise 4 - Grouped Numerical Data and Histograms

Invoice amounts are grouped into intervals:

Interval	Frequency
0 to 50	8
50 to 100	19
100 to 150	31
150 to 200	22
200 to 250	10

1. What is the sample size?
2. Which interval is most frequent?
3. If intervals are interpreted as $[0, 50)$, $[50, 100)$, ..., where is the value 50 counted?
4. Explain why grouped data lose information compared with raw data.
5. Explain how changing the bin width could change the appearance of a histogram.

Exercise 5 - Mean, Median, Mode, and Outliers

Use the dataset:

1, 2, 4, 5, 5, 5, 6, 6, 8, 9, 10.

1. Compute the mean, median, and mode.
2. Now add the observation 35 and recompute the mean, median, and mode.
3. Which measure changes the most?
4. Explain what this example teaches about robustness to extreme observations.

Exercise 6 - Mean from Tabulated Data

The number of items bought is summarized as follows:

Items bought	Frequency
1	4
2	9
3	15
4	11
5	6
6	5

1. Compute the sample size.
2. Compute the mean number of items bought.
3. Identify the mode.
4. Explain why this calculation is a weighted average.
5. Explain why simply averaging the displayed values 1, 2, 3, 4, 5, and 6 would be wrong.

Exercise 7 - Percentiles, Quartiles, and the IQR

The following values are already sorted:

3, 4, 5, 7, 8, 9, 10, 12, 15, 18, 20.

1. Identify the median.
2. Explain in words what the 25th percentile means.
3. Explain in words what the 75th percentile means.
4. Compute the interquartile range using $Q_1 = 5$ and $Q_3 = 15$.
5. Give one finance example where percentiles are more informative than the mean alone.

Exercise 8 - Dispersion and Risk

Two investments have the following five annual returns:

Investment A	Investment B
8%	-10%
9%	0%
10%	10%
11%	20%
12%	30%

1. Compute the mean return for each investment.
2. Compute the sample standard deviation (denominator $n - 1$) for each investment.
3. Compute the range for each investment.
4. Compute the coefficient of variation for each investment. Explain what it adds beyond the standard deviation, and state one situation in which the coefficient of variation would be misleading. (*stretch*)
5. Explain why the mean alone is not enough to describe risk.
6. Write a short investment interpretation using the words “location” and “dispersion.”

Exercise 9 - Variance Conventions and Python Defaults

Consider the values:

4, 6, 8, 10.

1. Compute the sample mean.
2. Compute the sum of squared deviations from the mean.
3. Compute the variance using denominator n .
4. Compute the variance using denominator $n - 1$.
5. Compute both corresponding standard deviations.
6. Which denominator is used by default in `numpy.std()`? Which is used by default in pandas `.std()`? Which parameter controls this?
7. Explain why the two answers differ, and when the difference is negligible.

Exercise 10 - Simple and Log Returns

An asset has the following price series:

Date	Price
0	100
1	102
2	101
3	103

1. Compute the simple return for dates 1, 2, and 3.
2. Compute the log return for dates 1, 2, and 3.
3. Explain why date 0 has no return.
4. Compare the simple and log returns. Are they close in this example?
5. Explain why returns are usually more comparable across assets than price changes.

Exercise 11 - Compounding Returns

Suppose an asset earns a simple return of 5% in the first period and -3% in the second period.

1. Compute the two-period return by compounding simple returns.
2. Compare the result with the simple sum $5\% - 3\%$.
3. Explain why the two numbers are not exactly the same.
4. Convert the two simple returns into log returns.
5. Show that the two-period log return is the sum of the one-period log returns.

Exercise 12 - Annualizing Return and Volatility

Suppose a stock has an average daily simple return of 0.08% and a daily volatility of 1.5%. Assume 252 trading days per year.

1. Compute the exact annualized average simple return using compounding.
2. Compute the linear approximation to the annualized average return.
3. Compare the two results. Is the gap larger or smaller than it would be for a smaller daily return? Explain briefly.
4. Compute the annualized volatility using the square-root-of-time rule.
5. Explain why volatility is multiplied by $\sqrt{252}$ and not by 252.
6. State one assumption behind the annualized volatility calculation.

Part 2 - Working with Python

Now move to real and larger datasets, where hand computation is no longer practical. Use `numpy`, `pandas`, and `matplotlib.pyplot`. The logic is the same as Part 1; only the scale and the tools change.

Before assuming any sheet or column name below, inspect the actual files first (for example with `pd.ExcelFile(path).sheet_names` and `df.columns`) and adjust your code to match the distributed data. This is good practice and avoids most “it does not run” errors.

Exercise 13 - Descriptive Statistics from the Workbook

Use the file:

`data and notebooks/Lecture 03 descriptive data.xlsx`

Use sheet `data1`, which contains one numerical variable called `Values`.

1. Load the sheet with `pd.read_excel()`.
2. Display the first rows and the shape of the dataset.
3. Compute count, mean, standard deviation, minimum, quartiles, and maximum.
4. Compute the interquartile range.
5. Draw a histogram.
6. Write a short paragraph describing the center, spread, and approximate shape of the distribution.

Exercise 14 - Categorical and Tabulated Data

Use the file:

`data and notebooks/Lecture 03 descriptive data.xlsx`

1. Load sheet `data3`, which contains an ordinal categorical variable.
2. Create a frequency table and a relative frequency table.
3. Draw a bar chart that keeps the categories in a meaningful order: `Very_Low`, `Low`, `Medium`, `High`, `Very_High`.
4. Identify the mode and write a short interpretation.
5. Load sheet `data6`, which contains exact integer values and their frequencies.
6. Compute the sample size, the weighted mean, and the mode from the tabulated data.
7. Explain why sheet `data6` is tabulated exact values, not grouped interval data, and why the weighted-mean formula from Exercise 6 applies here but not to the grouped data of Exercise 4.

Exercise 15 - Real Daily Prices and Returns

Use the file:

`data and notebooks/Lecture 03 - aapl daily 2012 2021.csv`

This file contains daily Apple prices with columns `Date`, `Open`, `High`, `Low`, `Close`, `Adj Close`, and `Volume`.

1. Load the CSV file.
2. Convert `Date` to a date column.
3. Sort the observations from oldest to newest.
4. Use `Adj Close` as the price series, and plot it as a line chart against time. Describe in one sentence what the *level* series shows before any transformation.
5. Compute daily simple returns with `.pct_change(fill_method=None)`.

6. Explain why the first return is missing.
7. Compute count, mean, standard deviation, minimum, quartiles, and maximum of daily returns.
8. Annualize the mean daily simple return using both the linear approximation and the compounded formula.
9. Annualize volatility using the square-root-of-time rule.
10. Draw a histogram of daily returns, and contrast what the line chart in step 4 and this histogram each reveal about the same data.
11. Write a contextualized descriptive statement. It should identify the asset, sample period, return definition, frequency, main descriptive statistics, and the limitation that these are historical sample statistics rather than forecasts.
12. (*stretch*) Sheet **data5** of the workbook **Lecture 03 descriptive data.xlsx** contains a second price series (a simulated price path beginning at 100). Treat it as a price series: compute its simple returns, summarize them (steps 5 and 7), and annualize its mean return and volatility (steps 8 and 9), assuming the same daily frequency. Then compare this series with Apple on annualized volatility. Which shows more dispersion, and does the higher average return come with the higher risk?

Exercise 16 - Two Numerical Variables and a Scatterplot

Sheet **data7** contains three continuous numerical variables, **X**, **Y**, and **Z**, measured on the same observations. Sheet **data8** contains two continuous numerical variables, **X** and **Y**.

1. Load **data7** and draw a scatterplot of **Y** against **X**. Describe the relationship in words: positive or negative, roughly linear or nonlinear, tight or loose, and note any visible outliers or clusters.
2. Draw a scatterplot of **Z** against **X** for the same sheet. How does the direction of this relationship differ from the first?
3. Compute the sample correlations between the pairs (for example with `.corr()`), and explain how the sign and magnitude of each correlation match what you saw in the scatterplots.
4. Now load **data8** and draw a scatterplot of **Y** against **X**, and compute the correlation. What does this dataset show that **data7** does not?
5. Explain why a scatterplot, rather than two separate histograms, is the natural first graph for a question about the *relationship* between two variables.

Exercise 17 - Summarize Every Sheet in the Workbook (*capstone*)

This exercise ties the whole lecture together. The workbook **Lecture 03 descriptive data.xlsx** contains several sheets, each holding a different kind of data.

1. List all sheet names programmatically.
2. Write a loop (or function) that, for each sheet, reports its shape, the variable name(s), and the pandas storage type (**dtype**) of each column.
3. For each variable, state its *statistical* type: nominal categorical, ordinal categorical, binary, discrete numerical, or continuous numerical. Note explicitly any case where the storage type and the statistical type disagree (recall Section 4 of the lecture: an integer column is not always a quantity to be averaged).
4. Produce the summaries that are *meaningful* for each type:
 - numerical variables: count, mean, standard deviation, minimum, quartiles, maximum;
 - categorical variables: a frequency table and the mode;
 - a tabulated sheet such as **data6**: the sample size, weighted mean, and mode computed from the value and frequency columns.
5. For at least two sheets, choose and draw an appropriate graph, and justify the choice.
6. Write one or two sentences of plain-language description for each sheet, as if documenting the dataset for a colleague who has not seen it.

The goal is a single, well-organized notebook section that a reader could use to understand every dataset in the workbook at a glance.